In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [4]:
```
df = pd.read_csv("Titanic-Dataset.csv")
```

In [5]:
```
df.head()
```

Out[5]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0

In [6]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    str    
 4   Sex          891 non-null    str    
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    str    
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str    
 11  Embarked     889 non-null    str    
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

In [7]:
```
df.describe()
```

Out[7]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [9]:
```
df["Age"] = df["Age"].fillna(df["Age"].median())
```

In [10]:
```
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
```

In [11]:
```
df.drop("Cabin", axis=1, inplace=True)
```

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

In [13]:
```
df.duplicated().sum()
```

Out[13]:
```
np.int64(0)
```

In [14]:
```
df["Sex"].value_counts()
```

Out[14]:
```
Sex
male      577
female    314
Name: count, dtype: int64
```

In [15]:
```
from sklearn.preprocessing import LabelEncoder

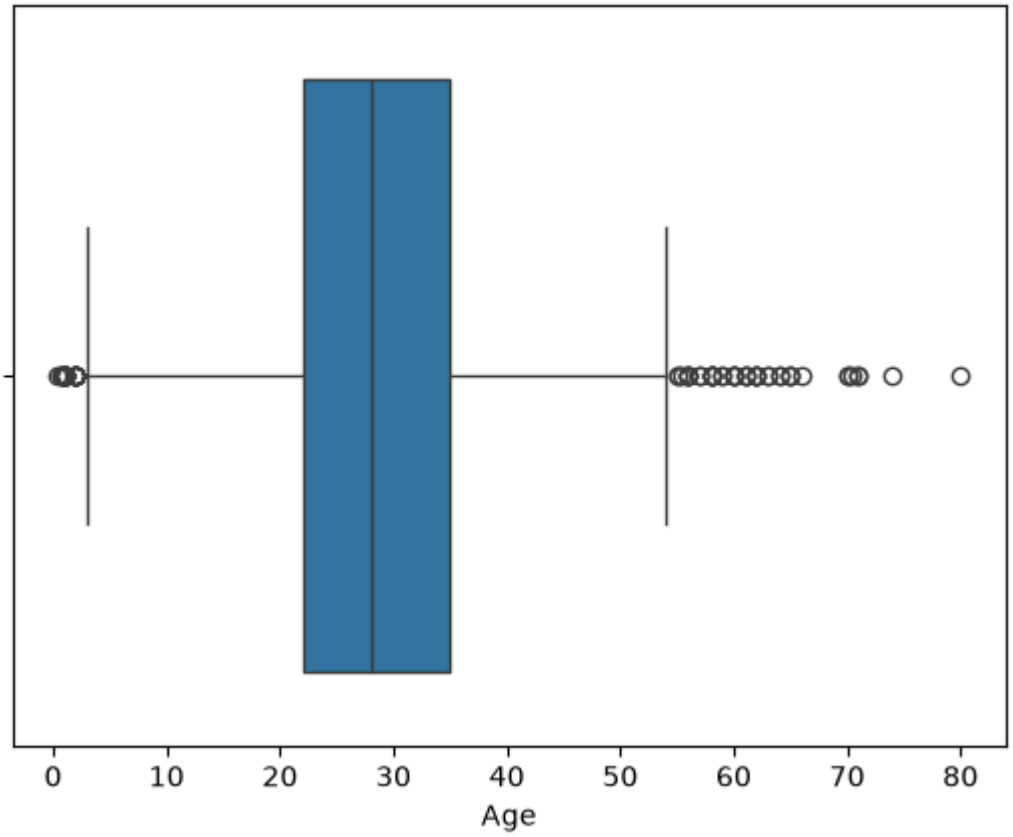
le = LabelEncoder()

df["Sex"] = le.fit_transform(df["Sex"])
```

In [16]:
```
df.select_dtypes(include=np.number).columns
```

Out[16]:
```
Index(['PassengerId', 'Survived', 'Pclass', 'Sex', 'Age', 'SibSp', 'Parch',
       'Fare'],
      dtype='str')
```

In [17]:
```
sns.boxplot(x=df["Age"])
plt.show()
```



In [18]:
```
Q1 = df["Age"].quantile(0.25)
Q3 = df["Age"].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)
```

```
Q1: 22.0
Q3: 35.0
IQR: 13.0
Lower Bound: 2.5
Upper Bound: 54.5
```

In [19]:
```
outliers = df[
    (df["Age"] < lower_bound) |
    (df["Age"] > upper_bound)
]

print(outliers)
```

```
     PassengerId  Survived  Pclass  \
7              8         0       3
11            12         1       1
15            16         1       2
16            17         0       3
33            34         0       2
..           ...       ...     ...
827          828         1       2
829          830         1       1
831          832         1       2
851          852         0       3
879          880         1       1

                                                Name  Sex    Age  SibSp  Parch  \
7                    Palsson, Master. Gosta Leonard    1   2.00      3      1
11                         Bonnell, Miss. Elizabeth    0  58.00      0      0
15                  Hewlett, Mrs. (Mary D Kingcome)    0  55.00      0      0
16                              Rice, Master. Eugene    1   2.00      4      1
33                             Wheadon, Mr. Edward H    1  66.00      0      0
..                                               ...  ...    ...    ...    ...
827                            Mallet, Master. Andre    1   1.00      0      2
829      Stone, Mrs. George Nelson (Martha Evelyn)    0  62.00      0      0
831                 Richards, Master. George Sibley    1   0.83      1      1
851                             Svensson, Mr. Johan    1  74.00      0      0
879  Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)    0  56.00      0      1

              Ticket     Fare Embarked
7             349909  21.0750        S
11            113783  26.5500        S
15            248706  16.0000        S
16            382652  29.1250        Q
33        C.A. 24579  10.5000        S
..               ...      ...      ...
827  S.C./PARIS 2079  37.0042        C
829           113572  80.0000        S
831            29106  18.7500        S
851           347060   7.7750        S
879            11767  83.1583        C

[66 rows x 11 columns]
```

In [20]:
```
df = df[
    (df["Age"] >= lower_bound) &
    (df["Age"] <= upper_bound)
]
```

In [21]:
```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df[["Fare"]] = scaler.fit_transform(df[["Fare"]])
```

In [22]:
```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

df[["Fare"]] = scaler.fit_transform(df[["Fare"]])
```

In [23]:
```
df["FamilySize"] = df["SibSp"] + df["Parch"] + 1
```

In [24]:
```
df[["SibSp", "Parch", "FamilySize"]].head()
```

Out[24]:

	SibSp	Parch	FamilySize
0	1	0	2
1	1	0	2
2	0	0	1
3	1	0	2
4	0	0	1

In [25]:
```
df["AgeGroup"] = pd.cut(
    df["Age"],
    bins=[0, 18, 60, 100],
    labels=["Child", "Adult", "Senior"]
)
```

In [26]:
```
df[["Age", "AgeGroup"]].head()
```

Out[26]:

	Age	AgeGroup
0	22.0	Adult
1	38.0	Adult
2	26.0	Adult
3	35.0	Adult
4	35.0	Adult

In [27]:
```
df.head()
```

Out[27]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	F
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	-0.485
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	38.0	1	0	PC 17599	0.797
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	-0.471
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	0.432
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	-0.469

In [28]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 825 entries, 0 to 890
Data columns (total 13 columns):
 #   Column       Non-Null Count  Dtype   
---  ------       --------------  -----   
 0   PassengerId  825 non-null    int64   
 1   Survived     825 non-null    int64   
 2   Pclass       825 non-null    int64   
 3   Name         825 non-null    str     
 4   Sex          825 non-null    int64   
 5   Age          825 non-null    float64 
 6   SibSp        825 non-null    int64   
 7   Parch        825 non-null    int64   
 8   Ticket       825 non-null    str     
 9   Fare         825 non-null    float64 
 10  Embarked     825 non-null    str     
 11  FamilySize   825 non-null    int64   
 12  AgeGroup     825 non-null    category
dtypes: category(1), float64(2), int64(7), str(3)
memory usage: 84.7 KB
```

In [29]:
```
df.describe()
```

Out[29]:

	PassengerId	Survived	Pclass	Sex	Age	SibSp	P
count	825.000000	825.000000	825.000000	825.000000	825.000000	825.000000	825.000
mean	447.369697	0.380606	2.341818	0.643636	28.515152	0.507879	0.357
std	257.088865	0.485830	0.824096	0.479215	10.177256	1.090670	0.798
min	1.000000	0.000000	1.000000	0.000000	3.000000	0.000000	0.000
25%	226.000000	0.000000	2.000000	0.000000	22.000000	0.000000	0.000
50%	445.000000	0.000000	3.000000	1.000000	28.000000	0.000000	0.000
75%	671.000000	1.000000	3.000000	1.000000	34.000000	1.000000	0.000
max	891.000000	1.000000	3.000000	1.000000	54.000000	8.000000	6.000

In [30]:
```
df.isnull().sum()
```

Out[30]:
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
FamilySize     0
AgeGroup       0
dtype: int64
```

In [31]:
```
df.duplicated().sum()
```

Out[31]:
```
np.int64(0)
```

In [32]:
```
df.to_csv("titanic_cleaned.csv", index=False)
```